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A coral reef is an underwater ecosystem formed as a ridge of mainly calcium carbonate deposits. Algae are photosynthesising organisms. Some algae grow on coral reefs. Succession results in a wide variety of fish living on coral reefs.

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1

Describe a method that could be used to determine the mean percentage cover of algae on a coral reef.

Do **not** include information on the difficulties of using your method underwater.

[3 marks]

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2

Explain how succession results in a wide variety of fish living on coral reefs.

Do **not** describe the process of succession in your answer.

[2 marks]

Ecologists investigated the effect of two fish species, the redband parrotfish and the ocean surgeonfish, on algal growth on an artificial reef. They made this artificial reef by submerging many large concrete blocks at a depth of 16–18 metres off the coast of Florida. They attached four large wire cages, **A**, **B**, **C** and **D**, to each block and populated the cages as shown.

A – No fish

B – Two redband parrotfish

C – Two ocean surgeonfish

D – One redband parrotfish and one ocean surgeonfish

After 34 weeks, the ecologists measured the mean percentage cover of all algae within each set of wire cages. The ecologists used a statistical test to find out whether the mean for each set of cages was significantly lower than the mean for set **A**.

Table 3 shows the probability (*P*) values that the ecologists obtained using this statistical test.

Table 3

Set of cages	<i>P</i> value
B	=0.841
C	<0.001
D	=0.634

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Using all the information, evaluate the effect of the two fish species on algal growth on coral reefs.

[5 marks]
